Naman Goyal

B.S. (Research) student in Physics, Indian Institute of Science, Bengaluru
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Education

Indian Institute of Science (IISc), Bengaluru Aug 2023 – Apr 2027 (expected)
B.S. (Research) in Physics; Cumulative GPA: 9.4/10

Modern Vidya Niketan (MVN), Faridabad, Haryana Apr 2020 – Apr 2023
Higher Secondary Schooling (CBSE); Class X: 97.8%, Class XII: 94.4%

Research Summary

My research interests lie in theoretical high energy physics and mathematical physics, focused on the fundamental constraints imposed by quantum field theory and symmetry principles. I combine analytical methods with scientific computing to study electroweak vacuum stability, semiclassical vacuum decay, scattering amplitudes, and effective descriptions of gravitation, with the aim of understanding the mathematical structure of fundamental interactions.

Publications & Preprints

[1] Naman Goyal, Banibrata Mukhopadhyay, and Ashish Kumar Meena. *Tests of general relativity in pseudo-Newtonian approach.* arXiv:2602.20261 [gr-qc], February 2026.

Research Experience

Scattering Amplitudes and S-Matrix Bootstrap May 2026 – Present
Prof. Rick Sandeepan Gupta, TIFR, Mumbai

- Investigating whether the uniqueness of dual-resonance scattering amplitudes can be established directly from analyticity, crossing symmetry, unitarity, and Regge behavior, avoiding auxiliary assumptions on their zero distribution.
- Studying analytic properties of scattering amplitudes: polynomial boundedness, positivity constraints, and the analytic structure in the complexified Mandelstam plane, drawing on modern developments in the S-matrix bootstrap program.

Higgs Vacuum Stability and the Cosmological Hierarchy Problem (SMVac) May 2024 – May 2026
Prof. Rick Sandeepan Gupta, TIFR, Mumbai

- Developed SMVac, a modular C++ framework for precision studies of electroweak vacuum stability: NNLO electroweak matching conditions, three-loop Standard Model beta functions for the gauge couplings (g_1, g_2, g_3), top Yukawa y_t, and Higgs quartic λ (RK4-evolved to the Planck scale), the one-loop Coleman-Weinberg effective potential, and semiclassical vacuum decay calculations.
- Computed Higgs vacuum decay rates from both the analytical Fubini-Lipatov formalism and a numerical RG-improved treatment that integrates the full running one-loop effective potential over the conformal bounce profile, with the Fubini-Lipatov scale optimized by golden-section search.
- Constructed stability phase diagrams in the (M_h, M_t) plane from million-point OpenMP-parallel parameter sweeps, characterizing where the numerical treatment deviates from the analytical approximation near the metastability boundary, and diagnosing the unphysical negative bounce actions of the strict Fubini-Lipatov ansatz (the inconsistency between the constant-quartic kinetic term and the running-coupling potential).
- Investigated the Cosmological Hierarchy Problem by modeling the Higgs field as a stochastic spectator during slow-roll inflation, studying whether volume-weighted anthropic selection during inflation can bias Higgs initial conditions toward the metastable electroweak vacuum without introducing new low-energy degrees of freedom.

Tests of General Relativity in Pseudo-Newtonian Frameworks Jan 2024 – May 2026
Prof. Banibrata Mukhopadhyay, IISc, Bengaluru

- Derived analytical expressions for perihelion precession, gravitational redshift, and gravitational lensing in several pseudo-Newtonian potentials, including the Paczyński-Wiita, Artemova-Björnsson-Novikov, and Mukhopadhyay potentials, and compared their predictions with Schwarzschild and Kerr general relativity.
- Developed a Python framework for the numerical analysis and visualization of orbital observables, generating the figures used to benchmark multiple pseudo-Newtonian models across different parameter regimes.
- Identified the regimes in which parameterized pseudo-Newtonian potentials successfully reproduce individual relativistic effects, while demonstrating that no single potential consistently captures perihelion precession, gravitational redshift, and gravitational lensing simultaneously. Published as arXiv:2602.20261 (2026).

Independent Reading: Lorentz Representations and the Little Group Formalism May 2025 – Jun 2025
Reading project supervised by Prof. Rick Sandeepan Gupta, TIFR, Mumbai

- Conducted a self-directed reading project on foundational quantum field theory, focusing on Lorentz and Poincaré group representations and the little-group formalism for massive and massless particles.

- Investigated continuous spin representations, studying the formalism developed by Schuster and Toro and analyzing its implications for exotic particle states beyond standard discrete spins.

Experimental Projects

Traveling Flame in Narrow Tubes 2024
International Physicists' Tournament (IPT), IISc, Bengaluru

- Designed and 3D-printed narrow tubes with controlled obstacles to study flame propagation dynamics for the International Physicists' Tournament.
- Conducted experiments characterizing ignition thresholds, flame speed regimes, and the transition from laminar deflagration to turbulent combustion.
- Analyzed high-speed footage and pressure data to quantify turbulence onset, providing insight into premixed flame behavior in confined geometries.

Jumping Hoop Phenomenon 2023
International Physicists' Tournament (IPT), IISc, Bengaluru

- Developed a theoretical model for the vertical jumping of a rigid hoop driven by an off-center sliding puck, incorporating rotational inertia, impulsive normal forces, and the shifting center of mass.
- Performed controlled experiments measuring jump heights and trajectories under varying mass ratios and initial conditions.
- Implemented numerical simulations in C++ to solve the coupled equations of motion, achieving good agreement with experimental observations.

Summer Schools & Academic Programs

Perimeter Institute Undergraduate Summer School (USS) Jun 2026
Perimeter Institute for Theoretical Physics, Waterloo, Canada

NIUS Physics Camp Jun 2024
Homi Bhabha Centre for Science Education (HBCSE-TIFR), Mumbai

Awards & Achievements

- **Joint Entrance Examination (JEE) 2023:** All India Rank 33 in JEE Advanced (∼180,000 candidates) and AIR 281 in JEE Main (∼1.1 million candidates).
- **PLANCKS 2024:** represented India at the IAPS PLANCKS finals in Dublin, Ireland (**11th-place global finish**); re-selected for the 2025 competition.
- **NIUS Physics Scholar (Batch 21.1):** selected by HBCSE-TIFR to a national program admitting approximately 70 undergraduates annually for advanced research training.
- **Indian Physics Tournament (IPT):** member of the IISc team, **National Runner-up** in 2023 and 2024, with contributions to theoretical modeling and experimental design.
- **KVPY Scholar:** awarded the KVPY fellowship with AIR 171 (SA stream), a national scholarship for research careers in the basic sciences.
- **Indian Astronomy Olympiad (InAO):** selected twice (2021, 2022) for the Orientation-cum-Selection Camp (OCSC), the final national selection stage for India's International Olympiad in Astronomy and Astrophysics team.
- **RMO & NSEP:** Top 1% state ranking in RMO (2020, 2021) and in NSEP (2020).

Relevant Coursework

Advanced theoretical physics: Quantum Field Theory I & II, Standard Model of Particle Physics, General Relativity, Advanced Mathematical Physics, Nuclear and Particle Physics. **Core physics & astrophysics:** Quantum Mechanics I & II, Classical Mechanics, Statistical Mechanics, Electrodynamics, Condensed Matter Physics, Thermal Physics, Fundamentals of Astrophysics. **Mathematics:** Linear Algebra and Analysis, Probability and Statistics, Mathematical Methods of Physics.

Technical Skills

Programming languages: C++, Python. **Scientific computing:** Adaptive ODE integration, multi-loop RGE evolution, numerical optimization, scientific visualization. **Libraries & software:** NumPy, SciPy, Matplotlib, SymPy, Mathematica. **Development tools:** Git, LaTeX.